

How Replicas Are Organised, and What AmberMeta Can Detect

Findings from the deposited simulation corpus

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1 Scope

This note accompanies the AmberMeta SOP and answers the standing question of how users organise replicas and whether AmberMeta can recognise them. It reports what was found in the deposited corpus at `/store7/gentile/data/simulations`, what AmberMeta does with each layout today, and what would have to change to cover the cases it declines.

No code was changed. Everything below was measured against AmberMeta v1.1.0 as released.

2 The corpus

Of 50 system directories, 19 hold simulation files. 30 are empty placeholders. One, `sys005`, is *unreadable*: its permissions are `drw-rw-r--`, so the execute bit is missing and no user (including its owner) can traverse it. That is a one-line fix at the source (`chmod u+rx`) and worth doing before any further deposits.

2.1 What was tested

Five systems were selected to cover every replica convention present. All were scanned read-only; nothing was written into the corpus.

System	Files	Logs	Steps	Scan	Layout
sys023	1044	205	213	6.4 s	Chunked, no replicas
sys011	1280	260	268	1.9 s	run1..run4 beside the stages
sys004	5100	0	107	83.5 s	run1..run6 inside each stage
sys021	5348	1091	1105	27.8 s	01..05 inside each stage
sys024	167	30	50	1.3 s	L12Y/prod/01..05

All five were parsed without error. Two results stand out independently of replicas:

- **sys004 contains no output logs at all.** 5100 files (2496 restarts, 2496 trajectories, 104 inputs) and not one `.mdout`. Less is lost than that sounds. The trajectories are NetCDF despite their `.mdcrd` names, and AmberMeta types a file by its content, so each run still reports its atom count, frame count, frame spacing, achieved simulation time, box type and volume statistics: one production chunk here reads 5200 to 6000 ps from the trajectory alone. What the missing logs cost is the temperature, the completion status, the wall clock, and the authoritative record of where each run began. The last of those has a consequence of its own, in §3.3.
- **sys011 has 260 logs but only 65 inputs.** The runs are recoverable from the logs, but the requested settings for most of them are not on record.

3 The four conventions, and what AmberMeta does

Convention	Shape	Auto	Manual
Replicas within stages	equilibration/run1..run6, production/run1..run8	yes	n/a
Numbered within stages	equil/01..05, prod/01..05	yes	n/a
Replicas beside stages	run1..run4 next to equilibration/equi1..equi4	no	yes
Id repeated in filenames	L12Y/prod/01/GB1_204_01_000	no	yes

Two of the four are recognised with no intervention, and produce correctly tagged members (run1..run8 for sys004; 01..05 for sys021). The other two are *declined*, not mis-grouped; in both, the runs are still discovered and still correctly chained. Only the grouping into members is withheld.

3.1 Why the two declines happen

Neither is a defect. AmberMeta's layout inference tags nothing when a tree admits more than one reading, on the grounds that a wrong member claim is worse than none.

sys011 presents two families whose tag sets are disjoint: equilibration/equi1..equi4, which varies at the second path segment, and run1..run4, which varies at the first. Disjoint families are refused because they are equally consistent with two separate experiments in one directory. Restricting the scan to the replica directories alone resolves it immediately: `discover --pattern 'run[0-9]' tags run1..run4` correctly.

sys024 repeats the replica identifier inside the filenames: prod/01/GB1_204_01_000, prod/02/GB1_204_02_000. Grouping keys on the set of run *bases* a directory holds, so each replica directory ends up with a base set of its own, no two directories are ever seen as running the same thing, and no cohort with more than one member is ever formed. This one cannot be fixed by narrowing the scan; the identifier has to be picked manually.

3.2 The manual route works for both

Define replicas... lets the user choose which path segment names the member. On sys011 choosing the first segment offers equilibration, leap, run1, run2, run3, run4; on sys024 choosing the third offers the preparation step names alongside 01..05. In both cases the replicas are present and correct among the options, and accepting only those produces a manifest identical to one from a conforming layout.

The cost is judgement: the picker offers every directory at the chosen level, including preparation directories that are not replicas. A user who accepts all of them gets a manifest asserting that leap is a replica.

3.3 A parser quirk this corpus exposed

sys004's production runs hold a trajectory and a restart and nothing else. Files are grouped by stem, so a run's own output restart lands in the slot continuity reads its *start* time from. v1.1.0 guards against exactly that, but the guard asks whether the group also holds an *mdin* or an *mdout*, on the reasoning that a group with neither is not a run at all but a bare topology and coordinate pair. Here both are absent and the runs are real, so the guard never fires: each chunk's own restart is read as its start time, and every junction reports a gap of one full chunk (999.9999998 ps against a 1000 ps chunk).

The equilibration directories carry *mdin* files, so the guard fires there and continuity reports an honest "cannot verify" instead. Over the whole system that splits cleanly: 2498 junctions report the cautious note, and 2399 report the phantom gap. A further 7 report an overlap of 299000 ps, which is the same misread start time landing on the far side of a chain boundary rather than inside one.

The condition is narrow, a deposit carrying neither inputs nor logs, but that is the shape a partial upload takes, and inside it the phantom gap the guard exists to prevent returns in full, on 2399 junctions of one system.

4 Recommendations

1. **Adopt the layout in §3 of the SOP for new work.** Replicas as sibling directories directly inside their stage, identically named across stages, with the identifier in the directory name only. Both auto-detected conventions in the corpus already follow this; it costs nothing to standardise on it.
2. **Treat missing output logs as a deposit blocker.** sys004 is the case to cite. The trajectories carry the timing, so the deposit is not worthless, but no run in it can be shown to have finished, no temperature is on record, and the continuity of every production chain is reported wrongly rather than cautiously (§3.3). A one-line check at deposit time, every run directory holding a *.mdout*, would have caught it.

3. **Raise the self-restart guard's condition as an issue.** Requiring an `mdin` or an `mdout` in the group is what makes it miss `sys004`. Keying on whether the group's coordinate file is also the run's declared output would cover the log-free case, and is worth a design pass rather than a quick patch.
4. **Fix `sys005`'s permissions** and re-run the survey over all 19 populated systems once the SOP is in use.
5. **Consider two contained improvements to the picker**, in order of value:
 - Mark which offered members *look like* replicas (directories whose run-base sets match each other) and which do not, so a user is steered away from accepting `leap` as a member. This is presentational and does not touch the inference rule.
 - Consider whether a directory whose base sets differ only by an embedded copy of its own directory name (the `sys024` shape) can be recognised safely. This is a real change to the membership predicate and would need its own design; the safe-failure property is worth more than the convenience, so it should not be rushed.
6. **Do not weaken the disjoint-families rule.** The `sys011` refusal is the rule working as intended. The procedural workaround is cheap and the failure mode it prevents, silently merging two experiments into one campaign, is expensive.